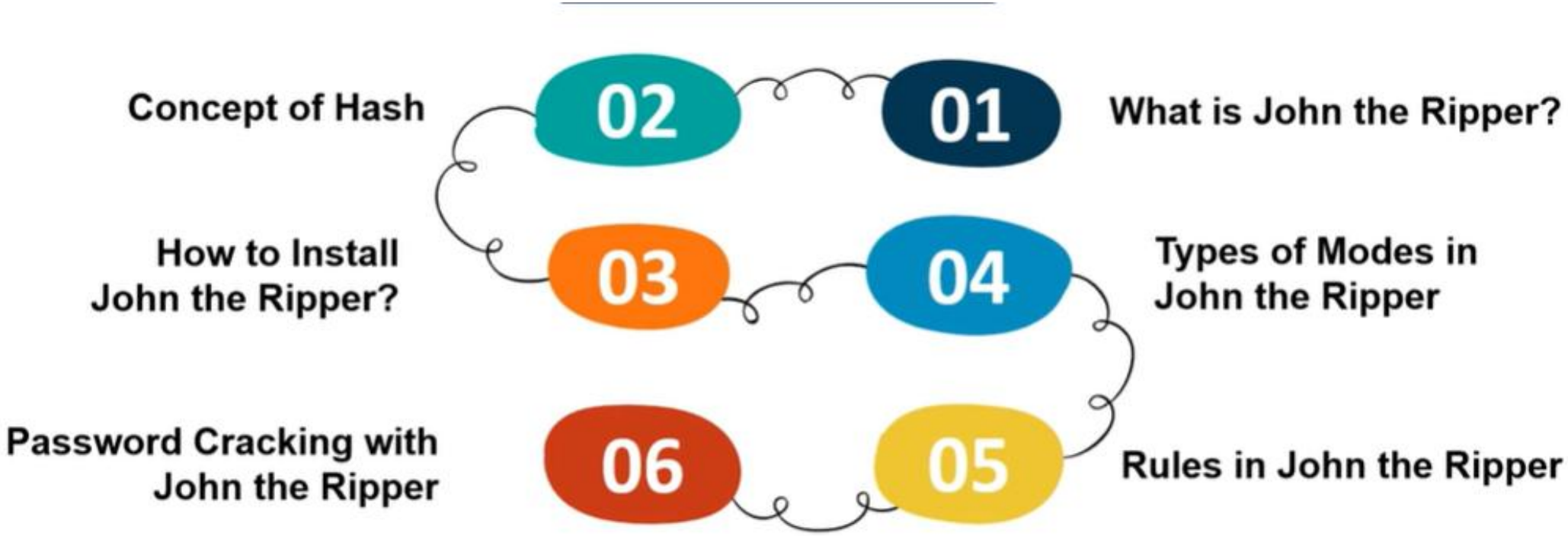
Password Cracking

TheJohnRipper

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What is John the Ripper?



John the Ripper is an Open-Source **Password Security Auditing** and **Password Recovery** Tool available for many **Operating Systems**.

- **Unix Flavors, macOS, Windows**
- **Web Apps**
- **Groupware**

- **Network Traffic Captures**
- **Encrypted Private Keys**
- **Filesystems & Disks**
- **Archives**

What is John the Ripper?

John the Ripper is an Open-Source **Password Security Auditing** and **Password Recovery** Tool available for many **Operating Systems**.



Security Experts



System

OR



File



Recover Passwords

What is John the Ripper?

John the Ripper is an Open-Source **Password Security Auditing** and **Password Recovery** Tool available for many **Operating Systems**.



Security Experts



Strength of Passwords

Strong Passwords

+

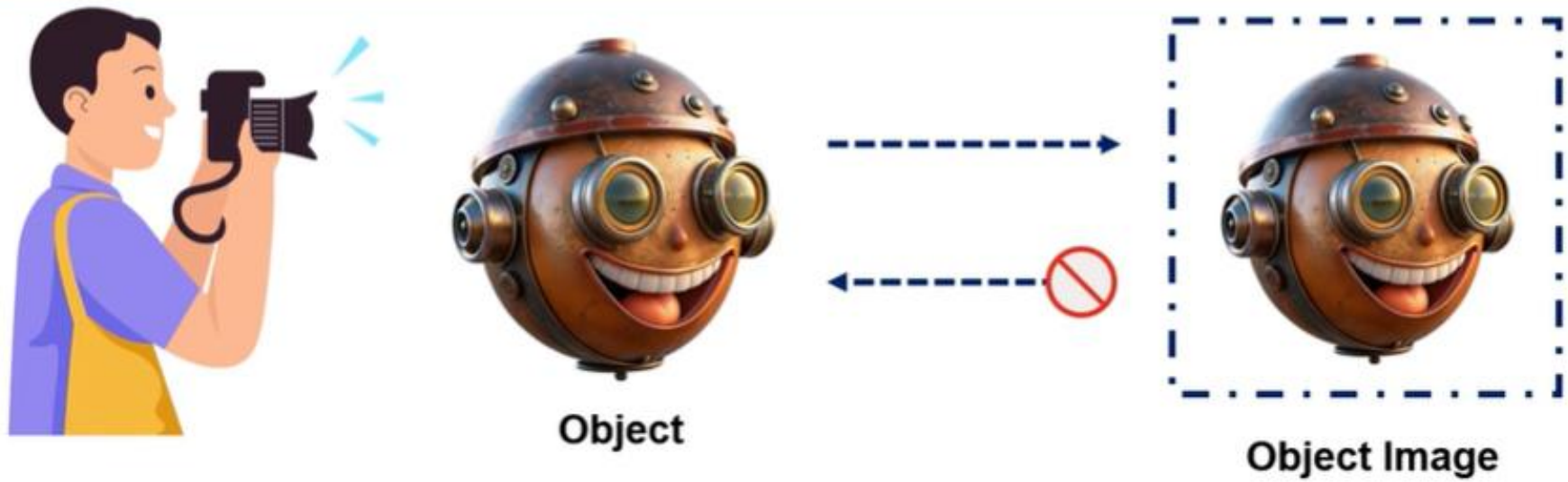
Weak Passwords



Improve Security

Concept of Hash

Hashing is a **Cryptographic** process that converts **Input Data** of any size into a **Fixed-Size String** of characters, known as a **Hash Value**.



Why use Hash for Password Storage?

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Why use Hash for Password Storage?

Hashing is a **Cryptographic** process that converts **Input Data** of any size into a **Fixed-Size String** of characters, known as a **Hash Value**.



Hacker

482c811da5d5b4bc6d497ffa98491e38

MD5 Hash



Recover Passwords

Commonly Used Hash Types

Hashing is a **Cryptographic** process that converts **Input Data** of any size into a **Fixed-Size String** of characters, known as a **Hash Value**.

MD5 (Message-Digest Algorithm 5)



MD5

482c811da5d5b4bc6d497ffa98491e38

Commonly Used Hash Types

Hashing is a **Cryptographic** process that converts **Input Data** of any size into a **Fixed-Size String** of characters, known as a **Hash Value**.

SHA-256 (Secure Hash Algorithm 256-bit)



SHA-256 → **ef92b778ba27b969d9656c6d29f4a5e89e36dfeeb252c56b275becc77c6b2759**

Commonly Used Hash Types

Hashing is a **Cryptographic** process that converts **Input Data** of any size into a **Fixed-Size String** of characters, known as a **Hash Value**.



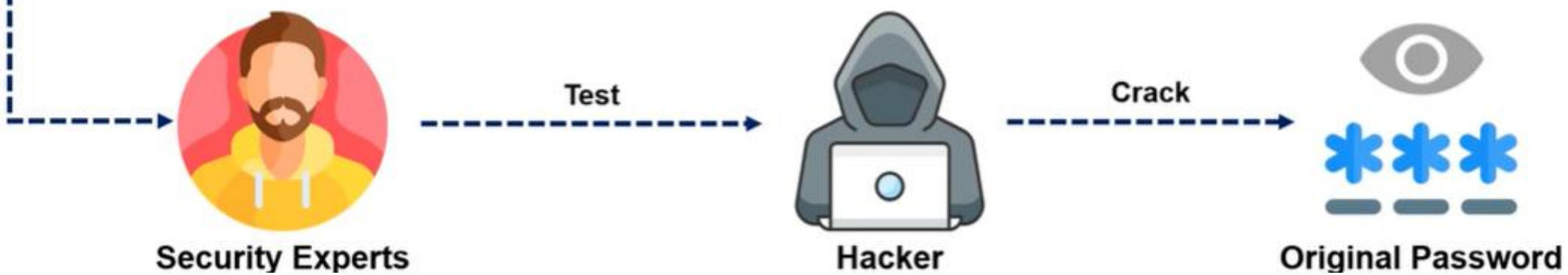
bcrypt

bcrypt

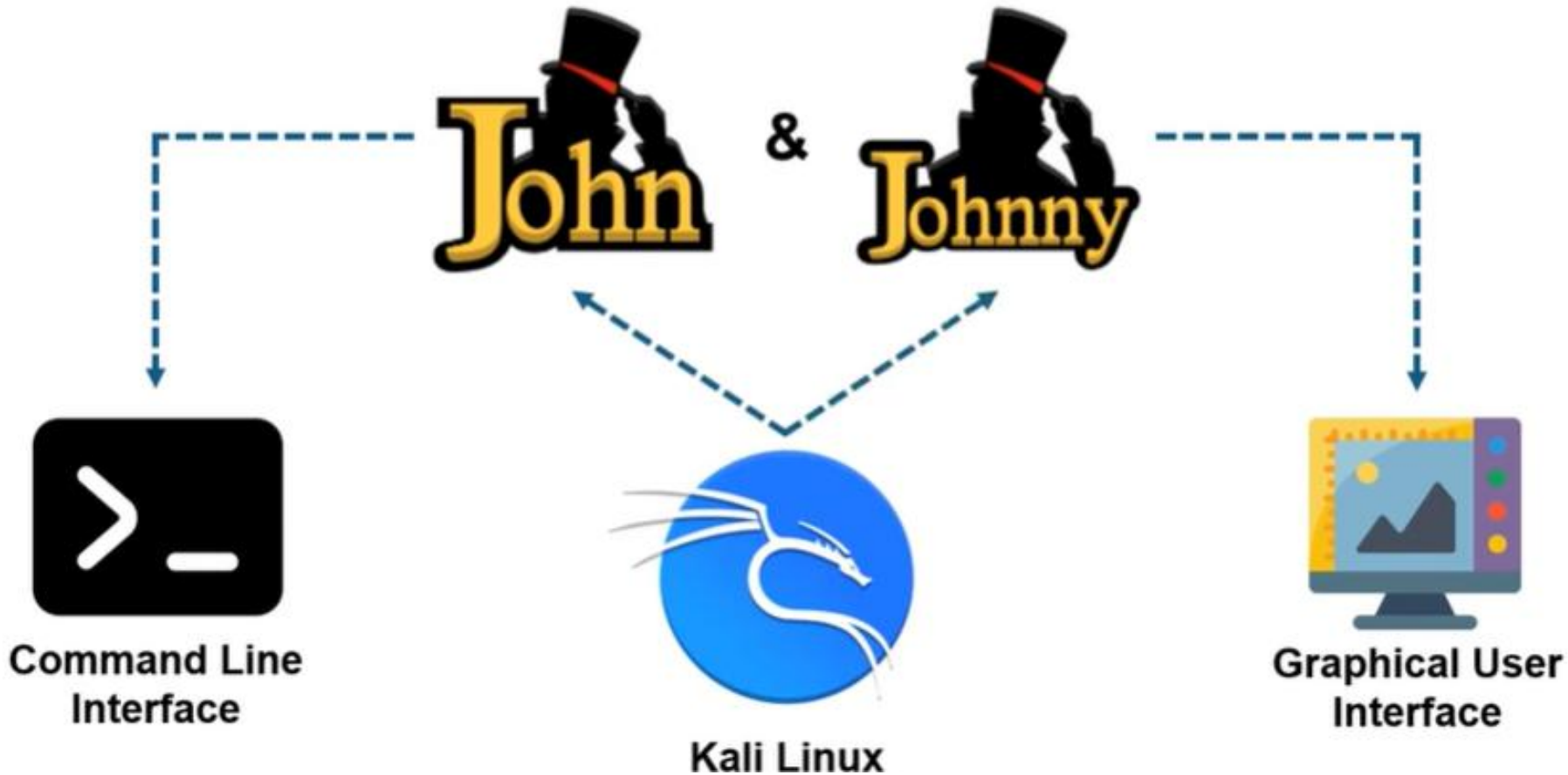
$2a$12$WXOeT6ezofQwXZRCMljM5ehfMQ1lgfjyhh/yFuSJwtaLWsdJI

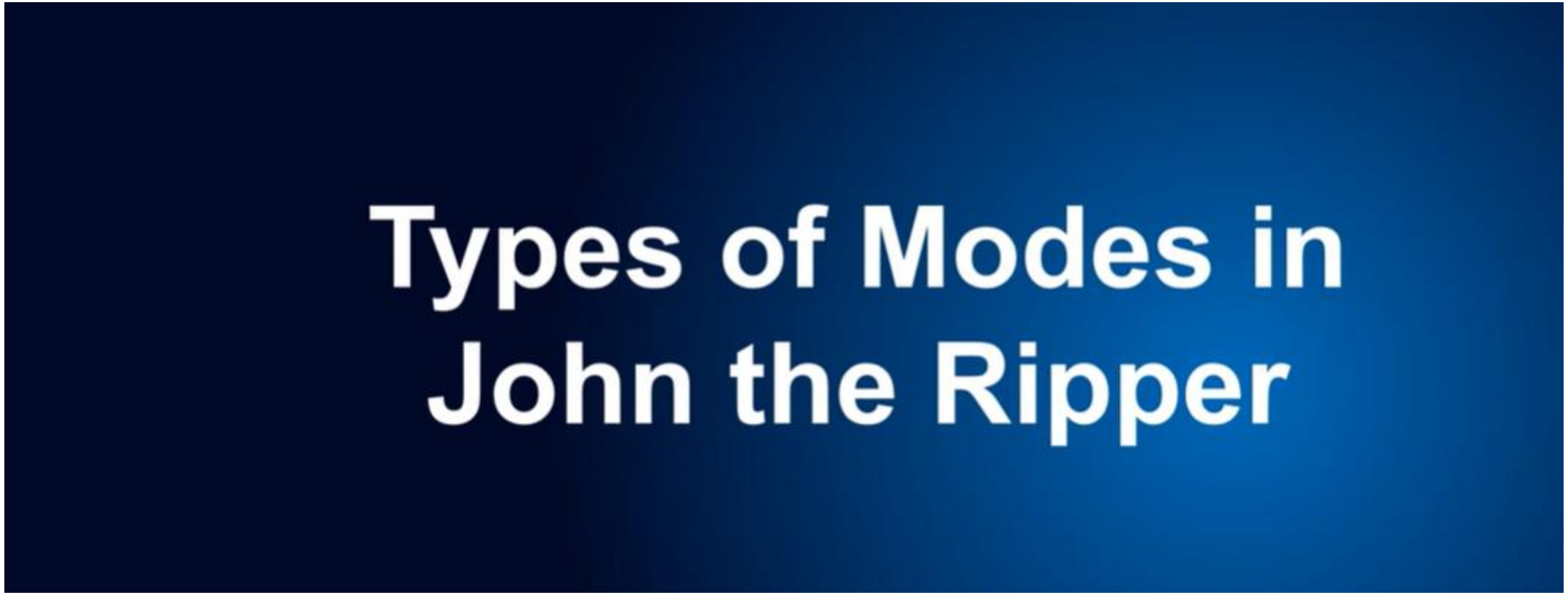
Why Cracking Hashes Important for Security Testing?

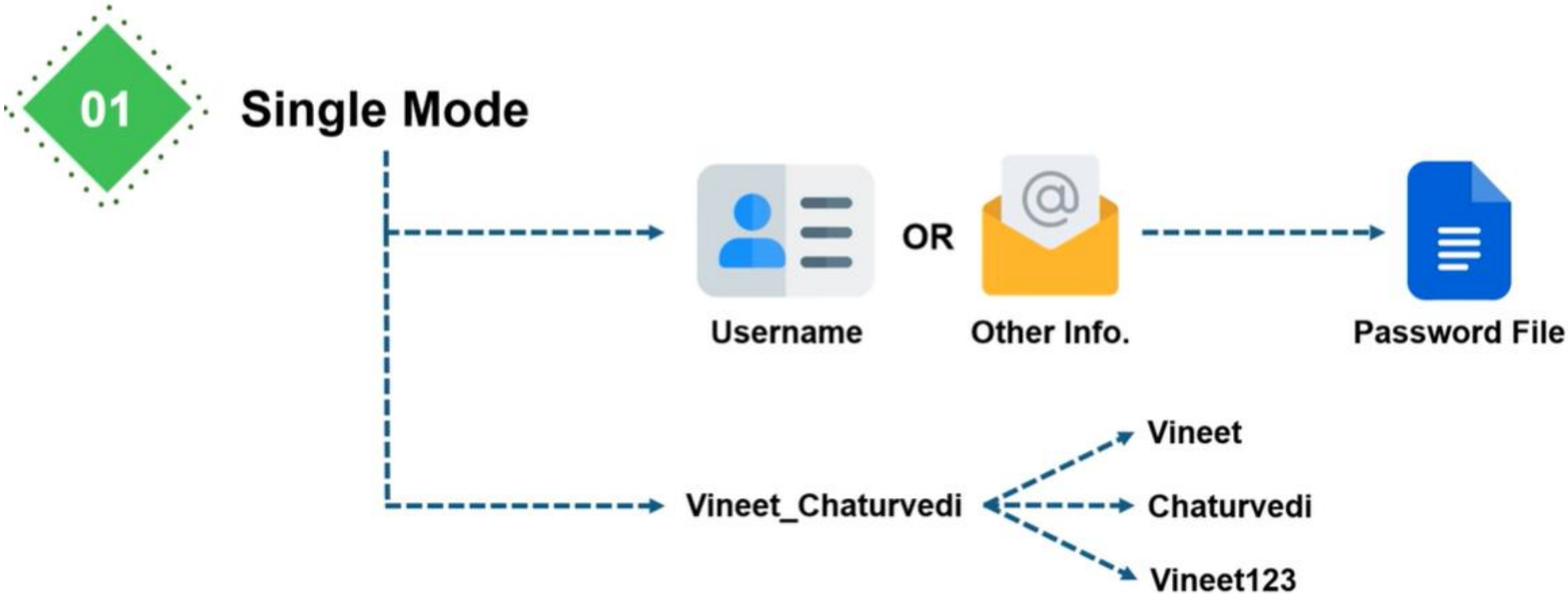
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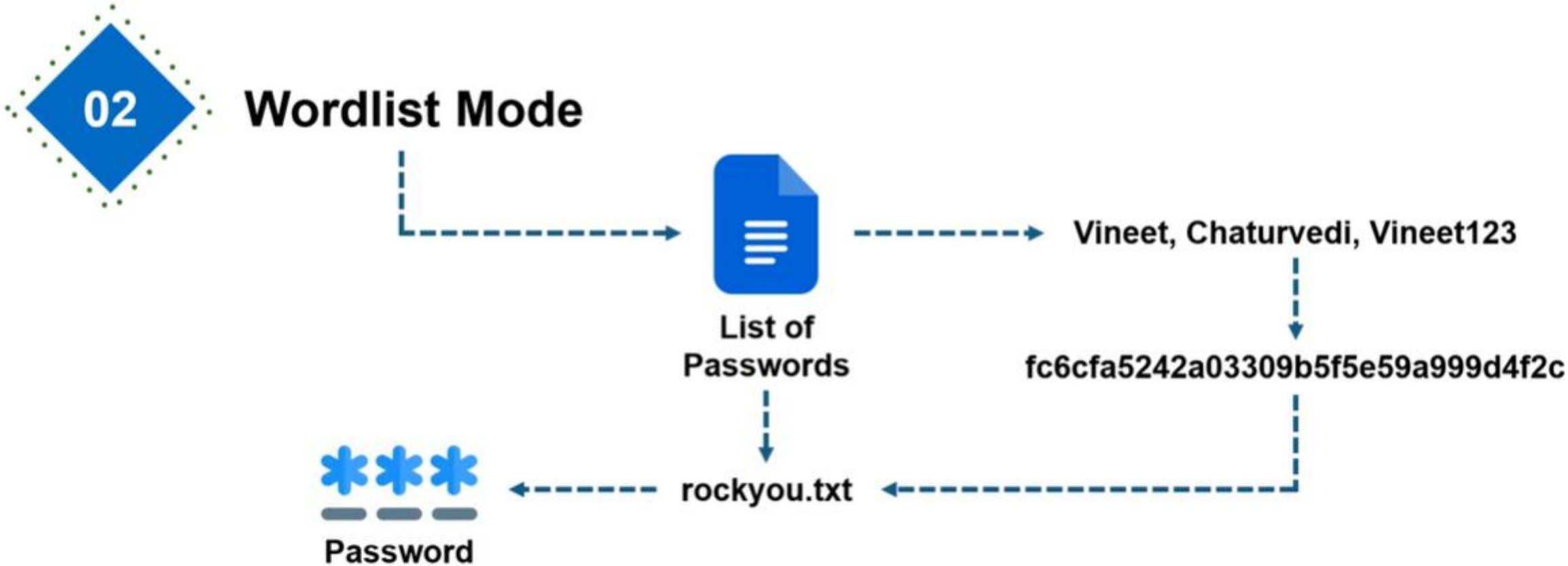


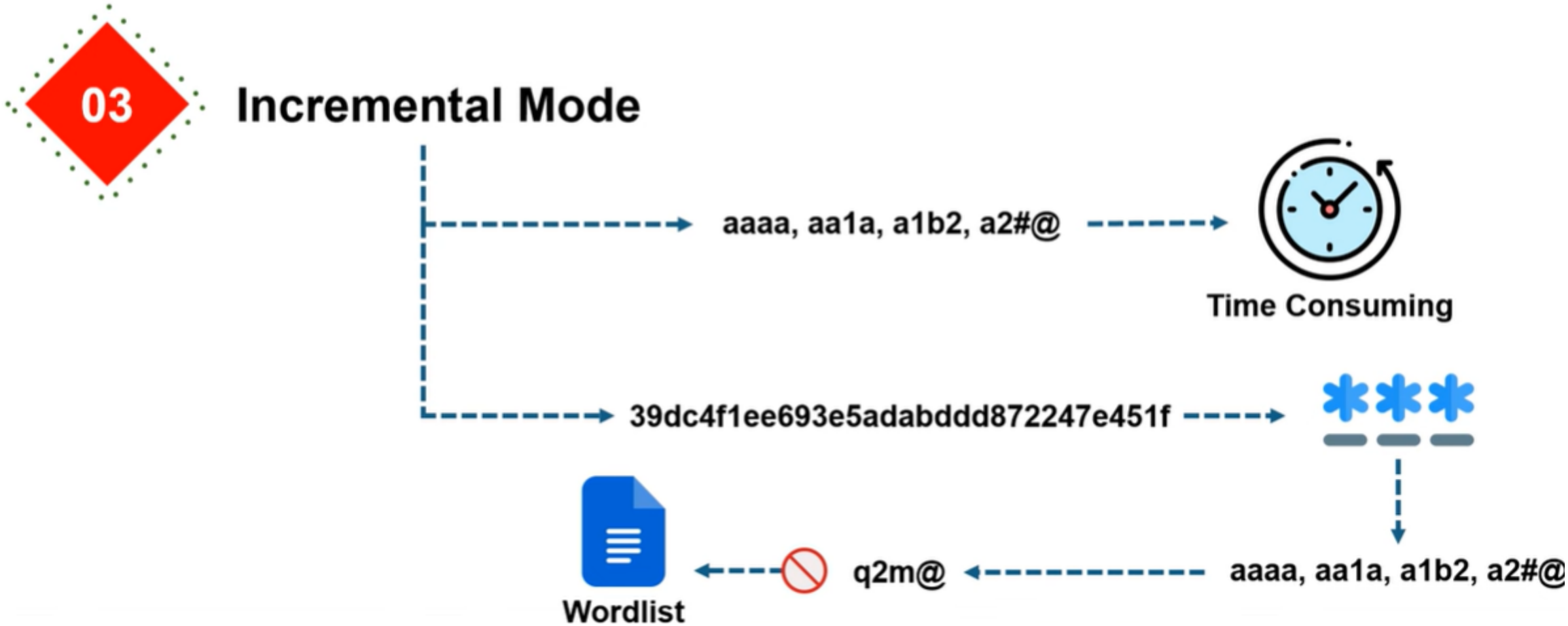
How to Install John the Ripper?




Types of Modes in
John the Ripper

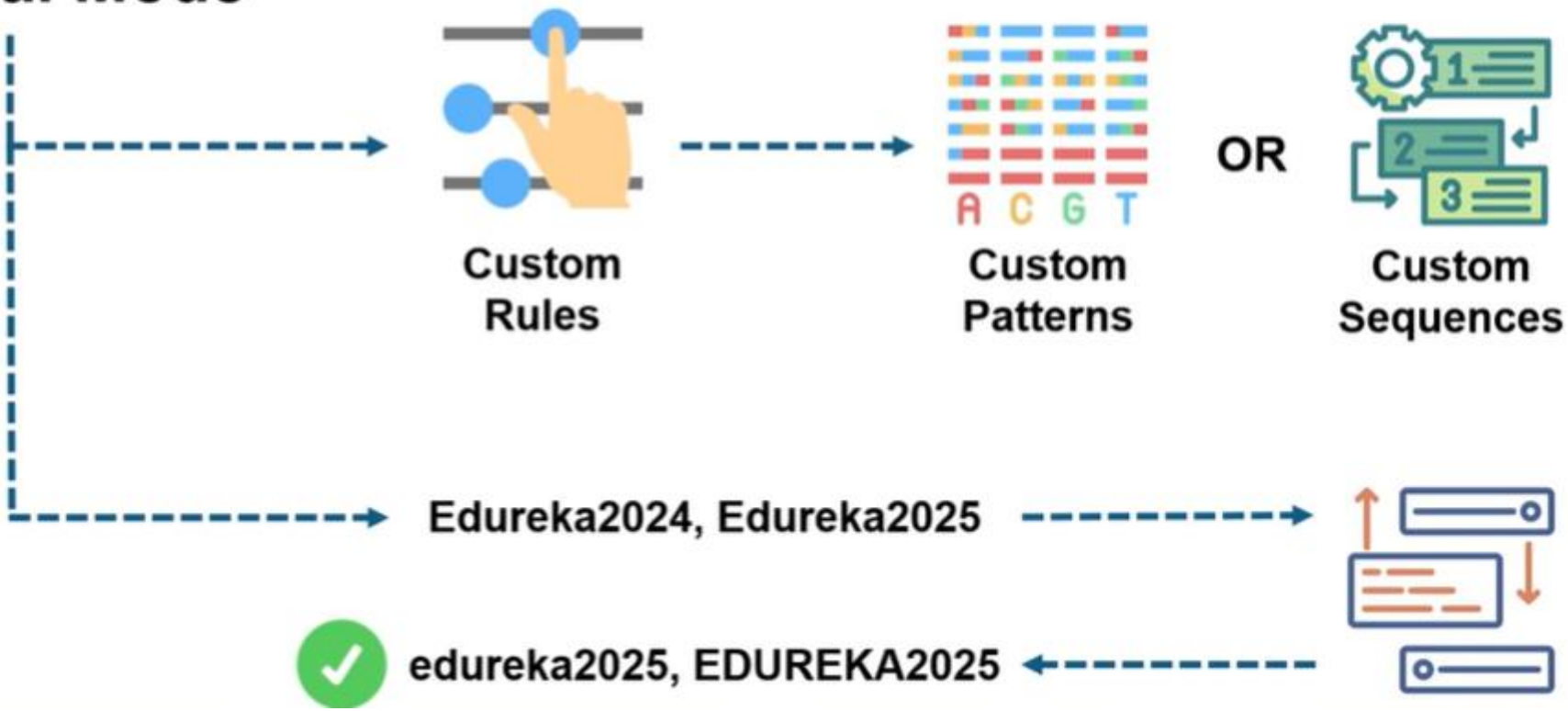

01
Single Mode
Username
OR
Other Info.
Password File
Vineet_Chaturvedi
Vineet
Chaturvedi
Vineet123


02
Wordlist Mode
List of
Passwords
Vineet, Chaturvedi, Vineet123
fc6cfa5242a03309b5f5e59a999d4f2c
rockyou.txt
Password

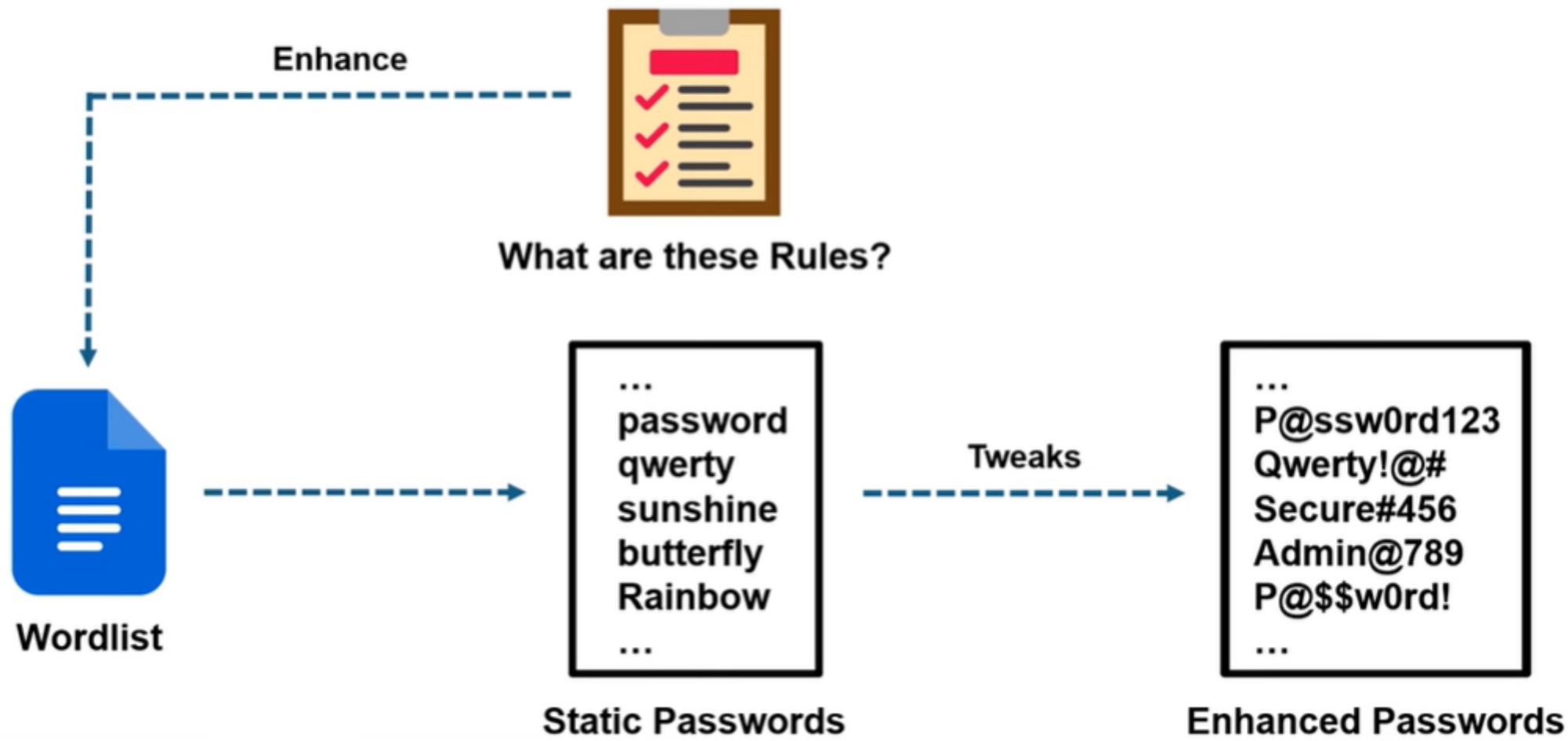

03
Incremental Mode
aaaa, aa1a, a1b2, a2#@
Time Consuming
39dc4f1ee693e5adabddd872247e451f
aaaa, aa1a, a1b2, a2#@
q2m@
Wordlist

04

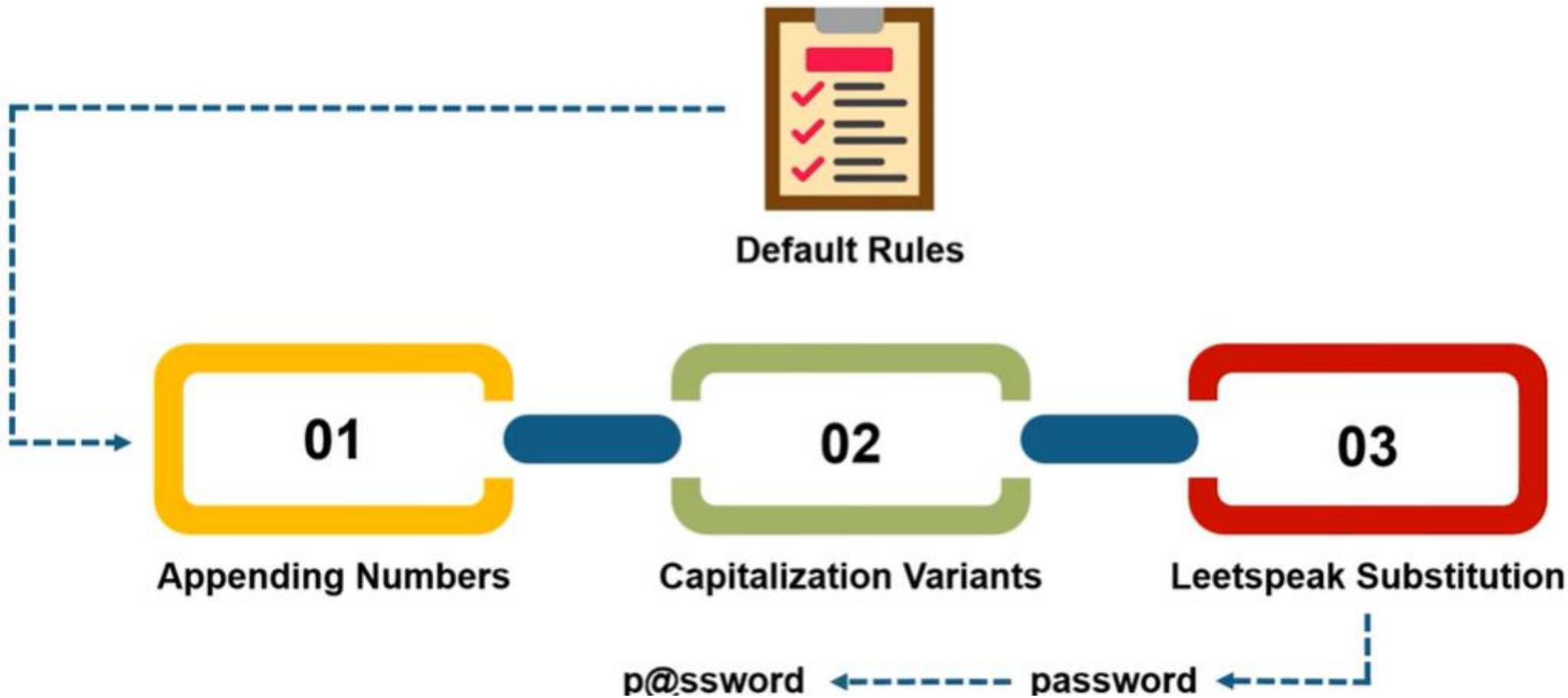
External Mode



Rules in John the Ripper



Rules in John the Ripper



Customizing Rules

Installing the John

- Checking
- john
- installing
- sudo apt install john
- Showing
- apt show john
- JOHN --HELP
- **GUI:**
- **sudoapt install johnny**
- **johnny**

Rules

- Sudo su-super file
- cd /etc/john-john directory
- nano john.confi

Search-list.Rules

[[A-a],[0-9]]

Why are Rules Important?



Rules



Birth Year/Common Suffix



Uppercase/Lowercase



Password Policy

Password Cracking with John the Ripper

- Create a hash file-touch hashfile.txt
- Wordlist file-touch wordlist.txt
- Locate file-cd /usr/share/wordlists
- Unzipfile-sudo gzip –d rockyou.txt.gz

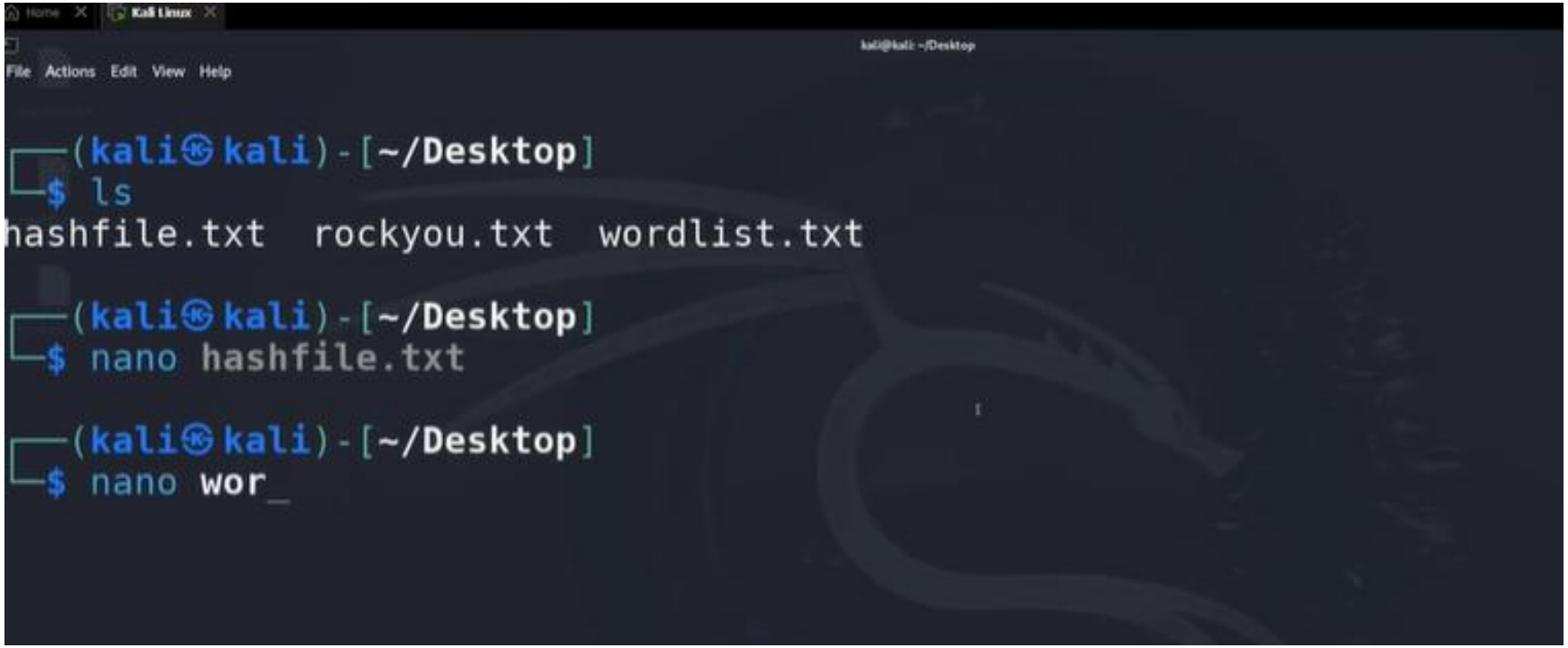
Supported Formats

- John –list=formats
- Basic syntax
- John hashfile.txt –format=RAW-MD5

Cracking Passwords using Single Mode

- Ls
- Cat hashfile.txt




Home
Kali Linux
kali@kali: ~/Desktop
File Actions Edit View Help
(kali㉿kali)-[~/Desktop]
$ ls
hashfile.txt rockyou.txt wordlist.txt
(kali㉿kali)-[~/Desktop]
$ nano hashfile.txt
(kali㉿kali)-[~/Desktop]
$ nano wor

Cracking Passwords using Wordlist Mode

JohnTheRipper

- John –wordlist=wordlist.txt hashfile.txt –format=RAW-MD5
- John –show hashfile.txt –formate=RAW-MD5
- Updated hashvalues-before move into incremental mode
- Ls
- Nano hash.txt
- Shift toggle rule
- Ls
- Nano wordlist.txt
- John –rules=

This is an Offline Password Hash Cracking process using John the Ripper

- **sudo su -**
- Switches to the **root** account (superuser). The - makes the shell a *login* shell so root's environment and home directory (/root) are used.
- You were then at the root㉿kali prompt.
- **cat /etc/passwd**
- Prints the content of /etc/passwd. Each line describes one user account in this format:
- username:password_placeholder:UID:GID:GECOS:home_directory:shell

PRACTICAL LAB – Password Cracking with John the Ripper (Kali Linux)

Modes: Single Mode, Wordlist Mode, Incremental Mode

- Part 1 — Create Test Users + Hash File
- 1. Switch to root
- sudo su -
- 2. Create two demo users
- adduser test01
- adduser test02
- 3. Copy /etc/shadow
- cp /etc/shadow hashfile.txt
- 4. Edit the file to include only test users
- nano hashfile.txt
- Part 2 — Single Mode Cracking
- john --single hashfile.txt
- john --show hashfile.txt
- Part 3 — Wordlist Mode Cracking
- sudo gzip -d /usr/share/wordlists/rockyou.txt.gz
- john --wordlist=/usr/share/wordlists/rockyou.txt hashfile.txt
- john --show hashfile.txt
- Part 4 — Incremental Mode (Brute Force)
- john --incremental hashfile.txt
- john --status
- john --show hashfile.txt
- Expected Results:
- Single Mode cracks weak passwords.
- Wordlist Mode cracks common passwords.
- Incremental Mode brute-forces short passwords.